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CELL BIOLOGY

# Generation of Stable Cell Lines of APP695 (*K595N, M596L*) Gene Over-Expression via Lentiviral Over-Expression System

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**Abstract**—There are now various theories on the pathophysiology of Alzheimer's disease (AD), the β-Amyloid (Aβ) cascade hypothesis has garnered a lot of interest. This text applied lentivirus-encapsulated plasmid to transfect APP695 (K595N, M596L) gene into SH-SY5Y cells and thereby obtain a stable cell line over-expressing Aβ. We used qPCR to monitor the mRNA expression of β-Amyloid Precursor Protein (APP) gene, β-Site APP Cleaving Enzyme 1 (BACE1) gene, we also used Western Blot to detect the APP protein, BACE1 protein, and $A\beta_{1-42}$, respectively, in addition, the expression of $A\beta_{1-42}$ in SH-SY5Y cell lysate and cell culture supernatant was measured by Enzyme-Linked Immunosorbent Assay (ELISA). The findings demonstrated APP gene over-expression following APP695 (K595N, M596L) gene transfection, and also indicated the successful establishment of the Swedish mutant cell model.

**Keywords:** Alzheimer's disease, stable cell lines, β-Amyloid, APP695 (*K595N*, *M596L*), BACE1

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## INTRODUCTION

AD is a typical neurodegenerative disease and the common cause of cognitive impairment in middle-aged and elderly people. It is estimated that 40 million people worldwide suffer from dementia, and this number should double every 20 years until 2050 (Yiannopoulou et al., 2020). AD is characterized by extracellular plaques containing Aβ and intracellular neurofibrillary tangles containing tau protein (Knopman et al., 2021). Aβ cascade hypothesis presents that deposition of Aβ plaques in brain is an early and important factor leading to loss of neurons in AD, and the deposition of Aβ can trigger a series of cascading pathological processes, which in turn further promote the deposition of Aβ (Wu et al., 2022; Vassar et al., 1999). Studies also point out that Aβ is the core of AD and that the deposition of Aβ is earlier than the formation of tau protein. Thus, suppressing Aβ production or enhancing the Aβ clearance draws much attention in the research field of AD treatment, many researchers have tried to find drugs for the treatment of AD based on such mechanisms (Wu et al., 2022; Lovestone et al., 2022), cell models with over-expressed Aβ therefore become essential for these researches.

Aβ peptide originates from APP through the amyloid metabolic pathway. APP is firstly cleaved by β-secretase to produce the 99-residue C-terminal fragment (C99), and then C99 is further cleaved by γ-secretase to produce Aβ peptide (Dehury et al., 2019; Haass C et al., 2012). The APP gene is found on the long arm of human chromosome 21st, and it produces several isoform proteins ranging in size from 100 to 130 kDa, among them, three subtypes of APP proteins—APP695, APP770, and APP751 can generate Aβ, and the most common APP subtype in the brain is APP695. Due to distinct cleavage sites of γ-secretase, different peptides of Aβ peptide are produced, with $A\beta_{1-40}$ and $A\beta_{1-42}$ being the more prevalent ones, $A\beta_{1-42}$ aggregates faster than $A\beta_{1-40}$ and is the major species in plaques (Yang et al., 2022). Herein, to construct a stable cell line with APP over-expression and thereby over-produce Aβ, we applied lentivirus-mediated transfection to transfect the plasmid with Swedish mutated APP gene (K595N/M596L; located in exon 16 of APP) into SH-SY5Y cells. The over-expression of APP was verified by qPCR and Western Blot. Moreover, BACE1, as one of the β-secretase and the most important factor for Aβ production was detected by qPCR and Western Blot, and the expression of $A\beta_{1-42}$

**Table 1.** Sequence of primers

| Gene | Primer sequences(5' → 3') | Amplified length |
|---|---|---|
| APP (Forward) | -TCGAGACACCTGGGGAT- | 125 |
| APP (Reverse) | -TGCTTGACGTTCTGCCT- | |
| BACE1 (Forward) | -GGCTTTCCAGGGTCTTTAC- | 200 |
| BACE1 (Reverse) | -TTCCTCTCCCCTCAACTTT- | |
| GAPDH (Forward) | -AAGGTCGGAGTCAACGGA- | 59 |
| GAPDH (Reverse) | -TTAAAAGCAGCCCTGGTGA- | |

was measured by ELISA and Western Blot. The establishment of the cell line in this paper can be applied to the research of AD pathogenesis and the development of innovative drugs under AD pathological conditions.

## MATERIALS AND METHODS

**Preparation of lentivirus.** The mammalian lentiviral expression vector GV358 was used to create the APP expression plasmid, and the Ubi-MCS-3FLAG-SV40-EGFP-IRES-puromycin sequence was put into the cloning site (mutations K595N and M596L in the coding sequence of APP transcript 201414). After transfecting HEK293T cells with the lentivirus-encapsulated $APP^{K595N/M596L}$ expression plasmid, the supernatant was harvested 48 h later, the lentivirus titer was determined to be $1 \times 10^9$ TU via qPCR. To create positive and negative cell populations, SH-SY5Y cells was transfected by lentiviruses carrying $APP^{K595N/M596L}$ /GFP and only GFP, respectively. Then, the homogenously transfected cells were collected and kept in a nutritional medium containing puromycin since all lentiviral plasmids conferred the resistance to puromycin. Meanwhile, SH-SY5Y cells without lentivirus transfection were used as a control.

**Cell culture and stable transfection.** SH-SY5Y cells were purchased from Shanghai Cell Bank of Chinese Academy of Sciences. They were cultured in high glucose DMEM (C11995500BT, Gibco) supplemented with 10% fetal bovine serum (C38010050, Gibco), 2 mmol/L glutamine (25030081, Sigma), 1% penicillin and streptomycin (SV30010, Hyclone). SH-SY5Y cells were planted at $3 \times 10^5$ cells/well into 6-well plates. When cell confluence reached 20%, medium-deprived FBS was added 2 h before transducing $APP^{K595N/M596L}$ lentiviral vectors with a multiplicity of infection (MOI) of 10 (MOI = 10). The medium containing lentivirus was discarded after 16 h and a new complete medium was added, GFP was observed under an inverted fluorescence microscope after 3 d of culture. Next, the complete medium with 2 μg/mL puromycin (MA0318, meilunbio) was added and cultured for 3 d to induce apoptosis of uninfected cells. Purified APPswe-SH-SY5Y cells were cultured in the medium with 20% FBS until the cells were confluent to the number they can grow stably. Finally, the complete medium containing 2 μg/mL puromycin was added to maintain a stable screening environment.

qPCR analysis of mRNA expression. SH-SY5Y cells were seeded in 6-well plates at a density of $1 \times 10^6$ cells/well, cultured in complete medium for 24 h, washed twice with PBS, and then the total cellular RNA was extracted with TRIzol reagent (15596026, Invitrogen), and the same amount of RNA was reversed-transcribed into cDNA with RT reagent Kit (RR037A). Afterwards, SYBR® Premix Ex Taq™ II (RR820A) was used to amplify cDNA and fluorescent signal was measured (qTOWER$^3$G, Analytik Jena). The method of $2^{(\Delta\Delta Ct)}$ was used to calculate the relative quantification of gene expression. The primer sequences are shown in Table 1.

Western Blot analysis of protein expression. SH-SY5Y cells were cultured in a 6-well plate at a density of $1 \times 10^6$ cells/well. After cultured in a complete medium for 24 h, the cells were detached via 0.25% trypsin-EDTA (G4001, Servicebio). After washing with PBS, 200 μL of RIPA lysate (PMSF was mixed in advance at a volume ratio of 100 : 1) (P0013K, Beyotime) was added and homogenized in ice for 30 minutes (gently mixed every 10 min), then cell lysate was centrifuged at 12000 *g* for 15 min at 4°C. The cell lysate was collected and protein concentration was measured by BCA kit (KGP902, KeyGEN), then mixed with 6× loading buffer (P1016, Solarbio), the sample was heated at the temperature of 100°C. The same amount of proteins was loaded onto 12% polyacrylamide gels and separated according to different molecular weights, then the separated proteins were electrotransferred to a PVDF membrane (IPVH00010, Millipore) and probed with primary antibodies against APP (PA1-12527, ThermoFisher), $A\beta_{1-42}$ (ab180956, Abcam), BACE1 (ab2077, Abcam), GAPDH (60004-1-Ig, Proteintech) and β-Actin (GB12001-100, Servicebio ) at 4°C overnight after blocking with 5% non-fat milk at room temperature with shaking on vortex. The membrane was washed three times by PBST and incubated with the secondary antibody, i.e. goat anti-mouse (SA00001-1, Proteintech) antibodies or goat anti-rabbit (SA00001-2, Proteintech) antibodies, at room temperature for 2 h on vortex. The PVDF membrane was washed 3 times with PBST and the protein band was detected with an enhanced ECL lumines-

cent solution and Western Blot detection system (UVP ChemStudio$^{TM}$815), quantitative analysis was performed via Image J software.

**Analysis of $A\beta_{1-42}$ content by ELISA.** SH-SY5Y cells were inoculated into 6-well plates at $6 \times 10^5$ cells/mL and cultured for 48 h, then they were lysed in RIPA lysis buffer (PMSF was mixed in advance at a volume ratio of 100 : 1). Meanwhile, the supernatant was centrifuged at 2000 *g* for 10 min, then a BCA assay kit was utilized to determine the concentration of total protein, $A\beta_{1-42}$ expression was detected in cell supernatant and lysates by using an $A\beta_{1-42}$ detection kit (CEA946Hu, cloud-clone), and the assay was performed according to manufacturer's instructions.

**TUNEL (terminal deoxynucleotidyl transferase-mediated dUTP-biotin nick end labeling assay) apoptosis detection.** SH-SY5Y cells were planted at $3 \times 10^5$ cells/well into 24-well plates and cultured for 24 h, the culture medium was discarded, the cells were washed three times with PBS, fixed with 4% paraformaldehyde, then TUNEL apoptosis detection was performed according to the instructions of the TUNEL kit's manufacturer's instructions (PF00009, Proteintech). Finally, nuclei were counterstained with DAPI, washed three times with PBS, and observed under an inverted fluorescence microscope.

Statistical analysis. All experiments were performed at least three times. Statistical evaluation was given via GraphPad Prism software, version 7.0. One-way ANOVA was used for comparison between two groups. All statistical results were obtained as $P < 0.05$, indicating that the difference was statistically significant.

## RESULTS

**Stable transduction.** SH-SY5Y cells grown in vitro are small in size, morphologically characterized as neuroblast-like, they often tend to grow in clusters and may form clumps as cells appear to grow on top of one another in the central region of a cell mass (Kovalevich et al., 2013). Bright image of SH-SY5Y cells before transfecting with the target gene is shown in Fig. 1a. Three experimental groups are set: the positive group represents that the cells are transfected with $APP^{K595N/M596L}$/GFP gene, the negative group represents that the cells are transfected with the GFP gene alone, and the control group represents the absence of lentivirus transfection. Green fluorescence was observed 3 d after SH-SY5Y cells were infected with lentivirus. Under the invert fluorescent microscope, it is visible that the cells of the negative group are small and show a typical shape of SH-SY5Y cells when compared to the control group. However, the cells of the positive group have a rounder shape and their antennae are shortened (Fig. 1b), and TUNEL staining shows that most of them are undergoing apoptosis because $A\beta_{1-42}$ deposition is neurotoxic and leads to the loss of neurons (Zhao et al., 2020). TUNEL stains the apoptotic nuclei in red, while DAPI stains all nuclei in blue (Fig. 1c). Notably, the negative control group displays weaker fluorescence than the positive group, in Fig. 1b, the number of bright green spots increased. Additionally, the swelled cells, the shortened antennae, and the irregular morphological changed cells, combined with the results of the TUNEL assay determined that the majority of the cells were undergoing apoptosis, therefore, there are more blebbing in the cells. And the expression of APPswe gene facilitates the synthesis of Aβ peptides, thus producing secretory vesicles. Above all, the blebbing and secretory vesicles can increase the reflection and further strengthen fluorescence density.

**Over-expression of APP.** Green fluorescence was observed to monitor transfection efficiency of the lentiviral plasmid. However, to further investigate whether the transfected APP gene can be expressed, we extracted the total RNA of cells, amplified the mRNA of APP gene and collected the fluorescence. simultaneously, we extracted total protein for Western Blotting. The expression of the APP protein was considerably higher in the cells of the positive group, as shown in Figs. 2a, 2b, RT-qPCR was carried out on the same quantity of mRNA in all groups. There is evidence showing that gene expression difference between the control group and the negative group is not immediately apparent, it can be observed that the gene expression of APP in the positive cells is elevated.

**Over-expression of $A\beta_{1-42}$.** Stable transfection of $APP^{K595N/M596L}$/GFP gene into cells allows over-expression of APP695 protein, but more research is needed to determine whether $A\beta_{1-42}$ over-expresses. We isolated proteins from culture supernatants and cell lysate respectively for Western Blot and ELISA. Of note, the concentration of $A\beta_{1-42}$ in the cell lysate is higher than that in the culture supernatant in the positive group (Fig. 3a), demonstrating that more $A\beta_{1-42}$ produced by the APP subtype of APP695 is retained in the cells rather than secreted from the cells. The results on ELISA reveal that in the cells of the positive group, both the cell lysate and the culture supernatant show a significant increase in the amount of $A\beta_{1-42}$, which is consistent with previous findings (Tao et al., 2019). Thereafter, we performed Western Blot to further detect $A\beta_{1-42}$ expression in the cell lysate. The outcomes were consistent with those of ELISA, a significant increase in the cells of the positive group was observed (Figs. 3b, 3c).

**Over-expression of BACE1.** Previous results show that APP695 transfection can construct a stable cell line with $A\beta_{1-42}$ over-expressed. Considering that BACE1 is a necessary β-secretase for Aβ production (Hampel et al., 2021), for the validity of the experimental results, we further used qPCR and Western Blot to detect the expression of BACE1. The qPCR results show that BACE1 gene is over-expressed in the cells of the positive group (Fig. 4a). Moreover, the

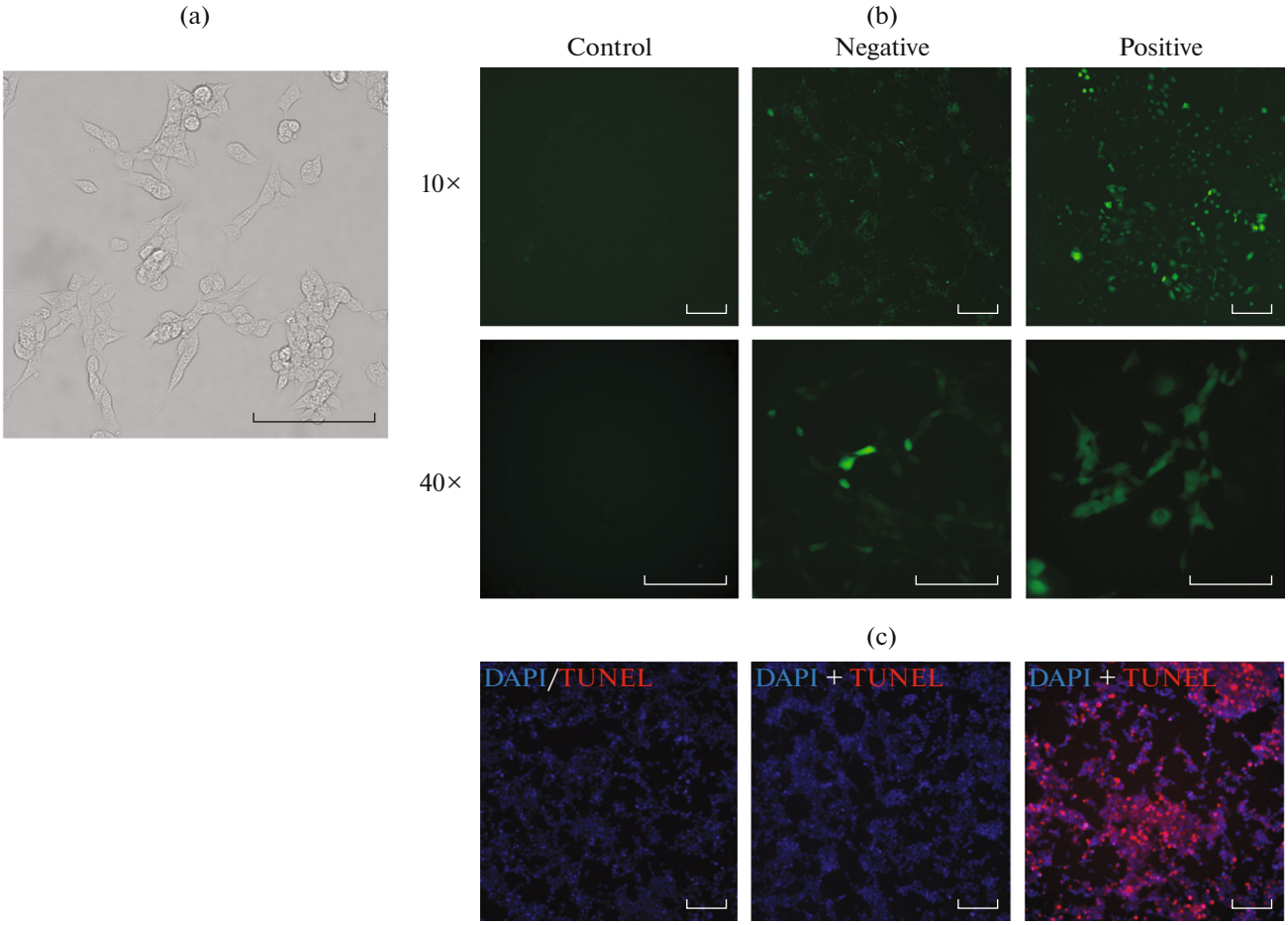


**Fig. 1.** Lentiviral infection efficiency. (a) Bright image of SH-SY5Y cells before transfecting with the target gene. Scale bar = 100 μm; (b) Fluorescence of SH-SY5Y cells after transfecting with the target gene. Scale bar = 100 μm; (c) TUNEL apoptosis detection to detect cell apoptosis, TUNEL stains the apoptotic nuclei in red, while DAPI stains all nuclei in blue. Scale bar = 100 μm.

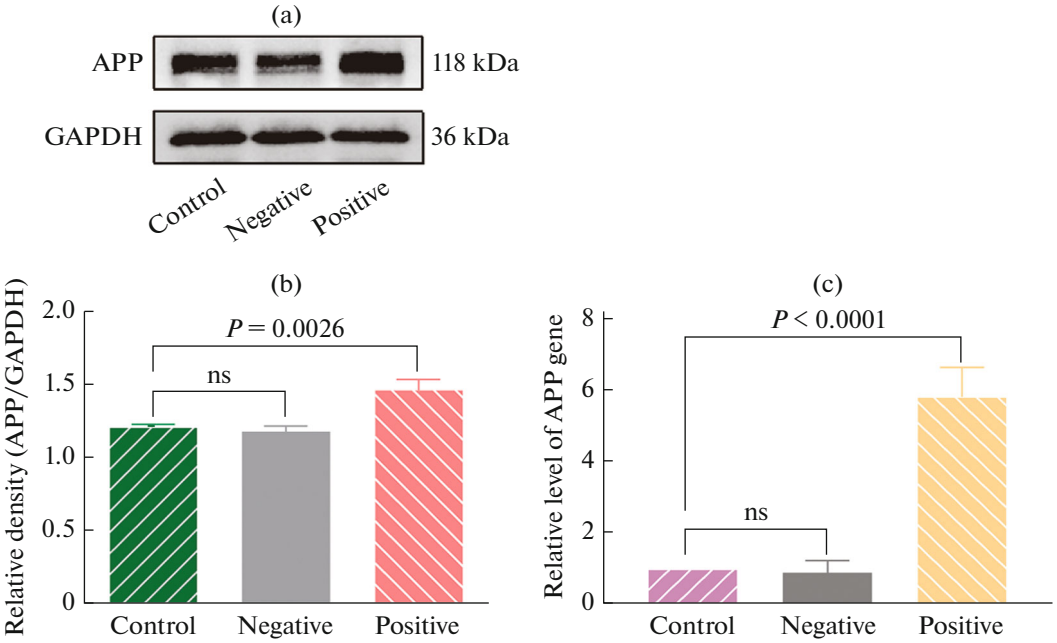


**Fig. 2.** Over-expression of APP gene. (a, b) The APP protein band by Western Blot ($P \leq 0.05$ refers to statistical difference, $n \geq 3$); (c) Relative expression of APP gene by RT-qPCR ($P \leq 0.05$ refers to statistical difference, $n \geq 3$).

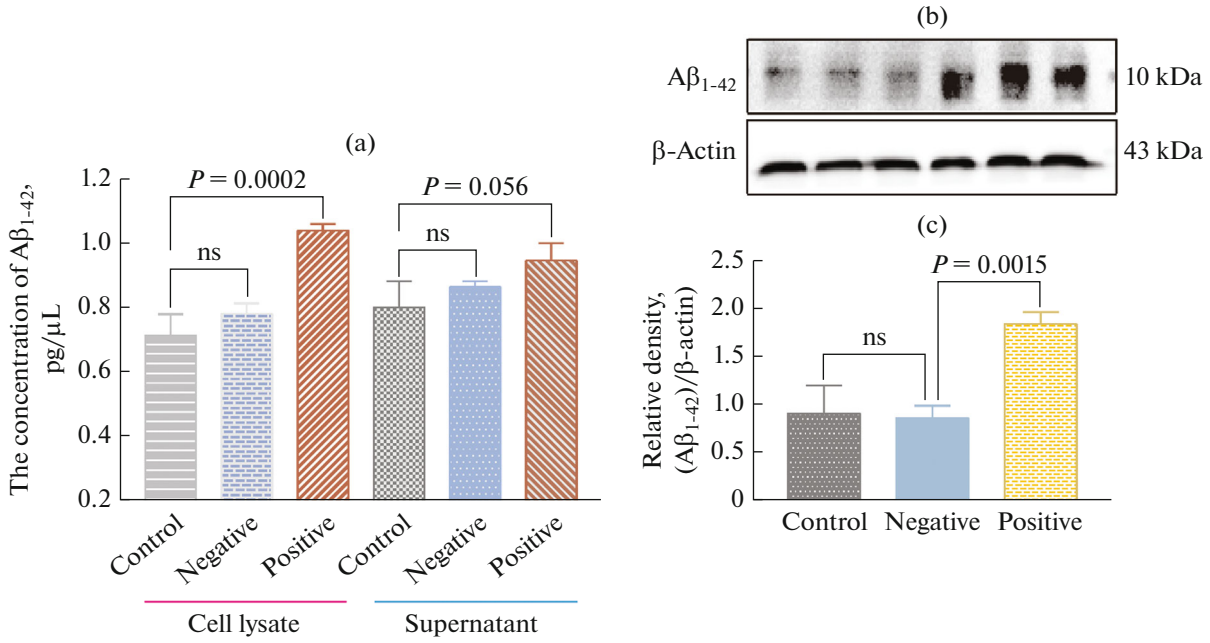


**Fig. 3.** Over-expression of $A\beta_{1-42}$. (a) Concentration of $A\beta_{1-42}$ in cell lysate and culture supernatant, respectively, obtained by ELISA ($P \leq 0.05$ presents statistical difference, $n \geq 3$); (b, c) Expression of $A\beta_{1-42}$ in cell lysate detected by Western Blot ($P \leq 0.05$ presents statistical difference, $n \geq 3$).

expression of BACE1 protein is also consistent with the results of qPCR (Figs. 4b, 4c).

## DISCUSSION

AD nowadays affects more than 40 million people worldwide and it is predicted to exponentially increase in the coming decades. Axonopathy including axonal swelling, impaired axonal transport, and synaptic dysfunction occurs early in AD patients (Stokin et al., 2005), and late-stage clinical manifests progressive memory loss and cognitive impairment. The main neuropathological features are the loss of nerve cells and a large number of senile plaques due to Aβ aggregation (Cheignon et al., 2018). Aβ is the core of AD and the deposition of Aβ is earlier than the formation of tau protein according to the Aβ cascade hypothesis though the likely link between Aβ aggregation and AD pathology needs a better understanding of the mechanisms. Accumulated evidence revealed that Aβ clearance can ameliorate cognitive deficits and as a promising therapeutic strategy (Conway et al., 2003; Wang et al., 2021), indeed, several potential pathways are involved in Aβ clearance, including phagocytosis and endocytosis by various cells, such as microglia, perivascular macrophages and astrocytes, proteolytic degradation by enzymes, including neprilysin, insulin-degrading enzyme and matrix metalloproteinases and other endogenous biomolecules, i.e. LRP1 and ApoE (Cheng et al., 2020; Van Gool et al., 2019; Liu et al., 2013), however, Aβ antibody-drug has not been ideal for the alleviation of cognitive impairment despite can removal of the Aβ peptide, so it remains a challenge to put these antibody-drug conjugates into clinical. Given the above consideration, blocking the generation of Aβ emerged to be a promising strategy for the therapy of AD. For this purpose, the Aβ over-expression cellular model needs to generate to clarify the mechanism of Aβ involving AD pathology, including neuroinflammation, toxicity, etc. In the onset of AD, Aβ is produced following APP shedding by β-(BACE1) and γ-secretase which is composed of PS1 or PS2 and nicastrin, Aph 1 and presenilin enhancer 2 (PEN2). The mutation of APP gene was discovered earlier compared to the presenilin (Watamura et al., 2022; Tao et al., 2019), and ApoE and tau protein were followed uncovered, so we over-expressed the Swedish mutant APP gene to generate the cellular model with Aβ overexpression.

SH-SY5Y is a neuroblastoma cell line, which is widely used in the study of neurodegenerative diseases, such as Parkinson's disease, AD, etc. Compared to primary cells, the overexpression of APP gene can exist stably for a long time, and avoid repeated transient transfection, saving cost and time. Currently, the animal models used in the field of Abeta include APP/PS1 transgenic mice, 5XFAD mice, etc., which employ mutant APP and presenilin 1 (PSEN1) genes, including the Swedish mutant of the APP gene, the Florida mutant, the London mutant, and the M146L and L286V mutations of the PSEN1 gene. Among them, this work uses the Swedish mutation of the APP

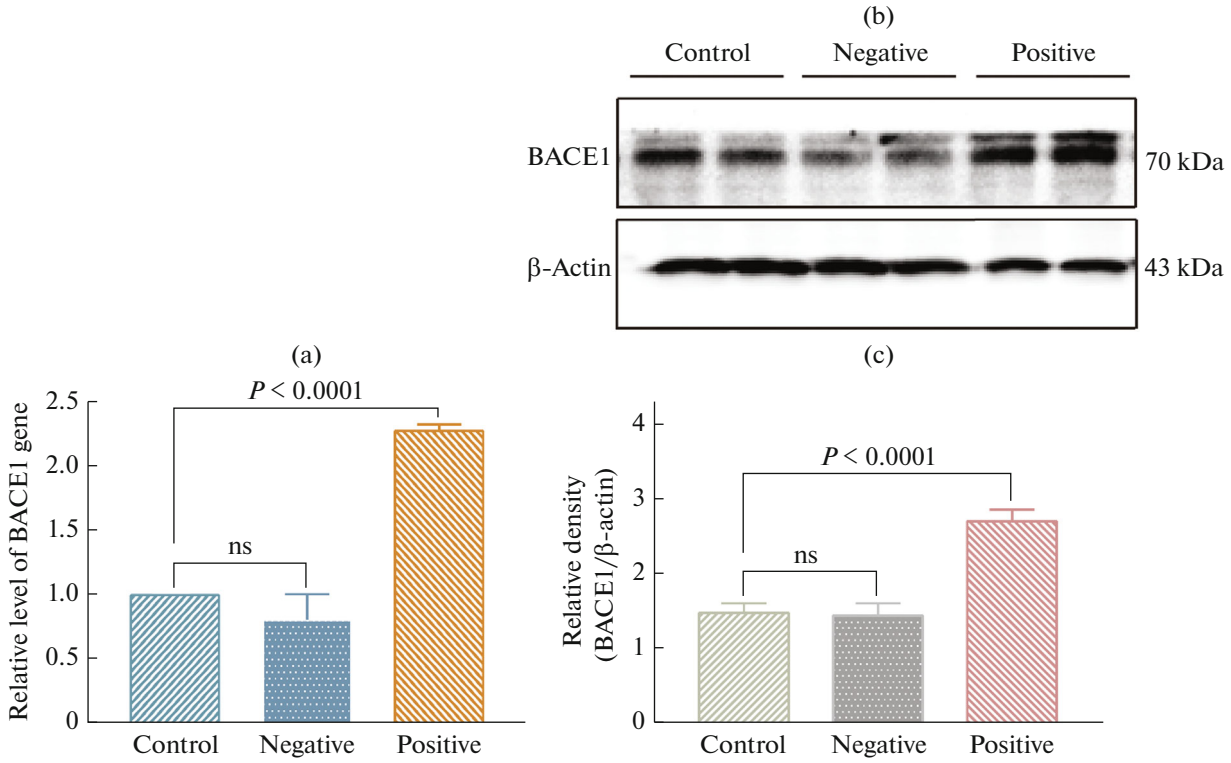


**Fig. 4.** Over-expression of BACE1. (a) The expression of BACE1 gene is detected by RT-qPCR ($P \leq 0.05$ presents statistical difference, $n \geq 3$); (b, c) BACE1 protein is measured by Western Blot ($P \leq 0.05$ presents statistical difference, $n \geq 3$).

gene to develop an Abeta overexpression cellular model, which has common aspects with animal models and complements each other from the perspective of research, one is in vitro model, and the other is in vivo animal model. In contrast, the establishment of cell line models saves time.

It has been reported there have several mutation sites of APP, the Swedish mutation APP (K595N/M596L) used in this text is more susceptible to cleavage by β-secretase to produce Aβ, so we construct the expression element Ubi-MCS-3FLAG-SV40-EGFP-IRES-puromycin and applied the GV358 vector to create APP overexpression plasmid.

## CONCLUSIONS

This paper generated a stable cell line with Aβ overexpression by over-expressing APP695 mutant gene in SH-SY5Y cells. Observation of GFP, staining on apoptotic cells, detection on the expression of APP, BACE1, and $A\beta_{1-42}$ were carried out successively. The expression status ultimately demonstrates successful construction of the stable cell line, the latter can be used for the study of the mechanism of Aβ deposition and the development of new AD therapeutic approaches against Aβ peptide.

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## AUTHOR CONTRIBUTIONS

Conceptualization: J. D., Y. Z. and H. Y.; Methodology: H. Z., R. H., J. D.; Formal analysis and investigation: H. Z., Z. Y., R. H., S. H.; Writing—original draft preparation: H. Z., J. R., S. W., H. S.; Writing—review and editing: J. D., Y. Z., H. Y., Y. H.; Funding acquisition: H. Y., P. Z.; Supervision: J. D., Y. Z. and H. Y.

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## ETHICS APPROVAL AND CONSENT TO PARTICIPATE

This work does not contain any studies involving human and animal subjects.

## CONFLICT OF INTEREST

The authors declare that they have no conflicts of interest.

## REFERENCES

Cheignon, C., Tomas, M., Bonnefont-Rousselot, D., et al., Oxidative stress and the amyloid beta peptide in Alzheimer's disease, *Redox Biol.*, 2018, vol. 14, pp. 450–464. https://doi.org/10.1016/j.redox.2017.10.014

Cheng, Y., Tian, D.Y., and Wang, Y.J., Peripheral clearance of brain-derived Aβ in Alzheimer's disease: pathophysiology and therapeutic perspectives, *Transl. Neurodegener.*, 2020, vol. 9, pp. 1–11. https://doi.org/10.1186/s40035-020-00195-1

Conway, K.A., Baxter, E.W., Felsenstein, K.M., et al., Emerging β-amyloid therapies for the treatment of Alzheimer's disease, *Curr. Pharm. Des.*, 2003, vol. 9, pp. 427–447. https://doi.org/10.2174/1381612033391649

Dehury, B., Tang, N., and Kepp, K.P., Molecular dynamics of C99-bound γ-secretase reveal two binding modes with distinct compactness, stability, and active-site retention: implications for Aβ production, *Biochem. J.*, 2019, vol. 476, pp. 1173–1189. https://doi.org/10.1042/bcj20190023

Haass, C., Kaether, C., Thinakaran, G., et al., Trafficking and proteolytic processing of APP, *Cold Spring Harbor Perspect. Med.*, 2012, vol. 2, p. a006270. https://doi.org/10.1101/cshperspect.a006270

Hampel, H., Vassar, R., De Strooper, B., et al., The β-secretase BACE1 in Alzheimer's disease, *Biol. Psychiatry*, 2021, vol. 89, pp. 745–756. https://doi.org/10.1016/j.biopsych.2020.02.001

Knopman, D.S., Amieva, H., Petersen, R.C., et al., Alzheimer disease, *Nat. Rev. Dis. Primers*, 2021, vol. 7, p. 33. https://doi.org/10.1038/s41572-021-00269-y

Kovalevich, J. and Langford, D., Considerations for the use of *SH-SY5Y* neuroblastoma cells in neurobiology, *Methods Mol. Biol.*, 2013, vol. 1078, pp. 9–21. https://doi.org/10.1007/978-1-62703-640-5_2

Liu, C.C., Kanekiyo, T., Xu, H., et al., Apolipoprotein E and Alzheimer disease: risk, mechanisms and therapy, *Nat. Rev. Neurol.*, 2013, vol. 9, pp. 106–118. https://doi.org/10.1038/nrneurol.2012.263

Lovestone, S., Fleshing out the amyloid cascade hypothesis: the molecular biology of Alzheimer's disease, *Dialogues Clin. Neurosci.*, 2022, vol. 2, p. 2000. https://doi.org/10.31887/dcns.2000.2.2/slovestone

Stokin, G.B., Lillo, C., Falzone, T.L., et al., Axonopathy and transport deficits early in the pathogenesis of Alzheimer's disease, *Science*, 2005, vol. 307, pp. 1282–1288. https://doi.org/10.1126/science.1105681

Tao, P., Zhao, Y., Song, X., et al., Preparation of the cells transfected with human *APPswe695* gene and the determination of Aβ production, *Chin. J. Cell Biol.*, 2019, vol. 41, pp. 72–79. http://dx.chinadoi.cn/10.11844/cjcb.2019.01.0008

Van Gool, B., Storck, S.E., Reekmans, S.M., et al., LRP1 has a predominant role in production over clearance of Aβ in a mouse model of Alzheimer's disease, *Mol. Neurobiol.*, 2019, vol. 56, pp. 7234–7245. https://doi.org/10.1007/s12035-019-1594-2

Vassar, R., Bennett, B.D., Babu-Khan, S., et al., β-Secretase cleavage of Alzheimer's amyloid precursor protein by the transmembrane aspartic protease BACE, *Science*, 1999, vol. 286, pp. 735–741. https://doi.org/10.1126/science.286.5440.735

Wang, H., Xu, X.X., Pan, Y.C., et al., Recognition and removal of Amyloid-β by a heteromultivalent macrocyclic coassembly: A potential strategy for the treatment of Alzheimer's Disease, *Adv. Mater.*, 2021, vol. 33, p. 2006483. https://doi.org/10.1002/adma.202006483

Watamura, N., Sato, K., and Saido, T.C., Mouse models of Alzheimer's disease for preclinical research, *Neurochem. Int.*, 2022, vol. 158, p. 105361. https://doi.org/10.1538/expanim.22-0164

Wu, T., Lin, D., Cheng, Y., et al., Amyloid cascade hypothesis for the treatment of Alzheimer's disease: progress and challenges, *Aging Dis.*, 2022, vol. 13, pp. 1745–1758. https://doi.org/10.14336/AD.2022.0412

Yang, Y., Arseni, D, Zhang, W., et al., Cryo-EM structures of amyloid-β42 filaments from human brains, *Science*, 2022, vol. 375, pp. 167–172. https://doi.org/10.1016/j.cclet.2022.06.046

Yiannopoulou, K.G. and Papageorgiou, S.G., Current and future treatments in Alzheimer disease: An update, *J. Cent. Nerv. Syst. Dis.*, 2020, vol. 12, pp. 1–12. https://doi.org/10.1177/1179573520907397

Zhao, J., Wang, M., Liu, W., et al., Activation of cannabinoid receptor 2 protects rat hippocampal neurons against Aβ-induced neuronal toxicity, *Neurosci. Lett.*, 2020, vol. 735, p. 135207. https://doi.org/10.1016/j.neulet.2020.135207

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